Homework — 1.3.1 Compression, Encryption and Hashing

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.3.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. A backup program stores software installers and a folder of source-code text files. State which type of compression (lossless or lossy) should be used and give **one** reason why this type is essential for these files. **[2]** *(AO1/AO2)*

2. Apply **Run Length Encoding** to the following row of pixels. Write your answer as value+count pairs. **[2]** *(AO2)*

`R R R R G G G R R B B B B`

3. A different row of pixels is `R G R G R G R G`. Explain why applying RLE to this row is a poor choice, referring to the size of the encoded output. **[2]** *(AO2)*

4. Explain how **dictionary-based compression** reduces the size of a large text document, and state **one** thing that must be supplied alongside the compressed data so it can be restored. **[3]** *(AO1/AO2)*

5. Alice wants to send a confidential file to Bob, whom she has never met, over the internet. Explain how **asymmetric encryption** allows Bob to receive the file so that only he can read it. Refer to the public key and the private key. **[3]** *(AO2)*

6. A website stores a hash of each user's password rather than the password itself.

(a) State **one** property of a hash function that makes this safe even if the database of hashes is stolen. **[1]** *(AO1)*

(b) Describe how the system checks that a user has entered the correct password at login, without ever storing the plaintext password. **[1]** *(AO2)*

Part B — Retrieval: keep it fresh (~6 marks)

7. *(1.2.4)* Explain **one** difference between a compiler and an interpreter. **[2]** *(AO1)*

8. *(1.1.1)* State what is held in the **MAR** and the **MDR** during the fetch stage of the fetch–decode–execute cycle. **[2]** *(AO1)*

9. *(1.2.1)* State the purpose of the **scheduler** in an operating system. **[2]** *(AO1)*